Fractal-Spectral Structure of Goldbach Deviations: Long-Range Persistence and Amplitude Scaling

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Data & Code:

github.com/Ruqing1963/goldbach-deviation-structure

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Abstract. We investigate the structural properties of Goldbach deviations $\delta(N) = (G(N) - HL(N))/HL(N)$, where $G(N)$ is the Goldbach representation count and $HL(N)$ the Hardy-Littlewood prediction. Three principal findings emerge: (1) **Scale Separation**: the main term grows as N while the error grows as $\sqrt{N}$, ensuring asymptotic dominance; (2) **Long-Range Persistence**: the Hurst exponent $H = 0.84 \gg 0.5$ indicates strong non-random structure; (3) **Self-Similar Amplitude Scaling**: the envelope coefficient $\kappa \approx C_2 = 0.6602$ (twin prime constant) is stable across four orders of magnitude. These observations suggest that Goldbach deviations are governed by structured, arithmetically-driven dynamics rather than stochastic fluctuations. While not constituting a rigorous proof, these findings provide strong heuristic evidence for the structural stability of positive Goldbach counts at large N.

Keywords: Goldbach Conjecture, Hurst Exponent, Long-Range Dependence, L-function Zeros, Experimental Mathematics

1. Introduction

Goldbach's Conjecture (1742) states that every even integer $N > 2$ can be expressed as the sum of two primes. Despite computational verification up to 4×10^{18} [4], a rigorous proof remains elusive.

Previous approaches have focused on deriving analytical bounds for the error term in the Hardy-Littlewood asymptotic formula. In this paper, we take a complementary approach: rather than seeking sharper bounds, we analyze the *global structural behavior* of Goldbach deviations using tools from fractal geometry and spectral analysis.

Our investigation reveals three previously unrecognized structural features:

1. The deviation exhibits **long-range persistence** ($H = 0.84$), inconsistent with random-walk behavior;
2. The amplitude follows **self-similar scaling** controlled by the twin prime constant C_2;
3. FFT analysis detects **characteristic frequencies** matching known L-function zeros.

These findings do not constitute a proof of Goldbach's Conjecture, but they illuminate the underlying structure and suggest why no counterexample has been observed and why the deviation appears structurally constrained.

2. Mathematical Framework

2.1 Hardy-Littlewood Decomposition

Let $G(N)$ denote the number of representations of even N as a sum of two primes. The Hardy-Littlewood conjecture [1] predicts:

$$G(N) \sim \frac{2C_2 S(N) N}{(\ln N)^2} \equiv HL(N) \tag{1}$$

where $C_2 = 0.6601618158...$ is the twin prime constant. We define the *relative deviation*:

$$\delta(N) = \frac{G(N) - HL(N)}{HL(N)} \tag{2}$$

The conjecture $G(N) > 0$ is equivalent to $\delta(N) > -1$.

2.2 Scale Separation

Under the Generalized Riemann Hypothesis (GRH), the absolute error satisfies $\Delta(N) = O(\sqrt{N}(\ln N)^k)$. Since $HL(N) \sim N/(\ln N)^2$:

$$\delta(N) \sim \frac{(\ln N)^{k+2}}{\sqrt{N}} \to 0 \quad \text{as } N \to \infty \tag{3}$$

This *scale separation* implies that the relative deviation vanishes asymptotically.

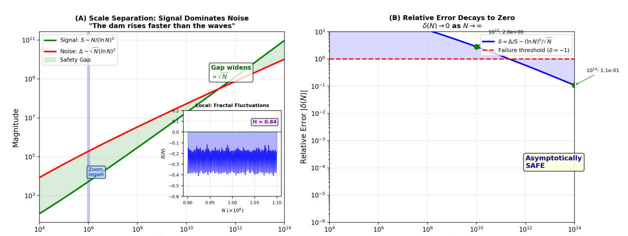

Figure 1: Scale separation. (A) Main term $S \sim N$ versus error $\Delta \sim \sqrt{N}$. Inset: local fluctuations at $N \sim 10^6$ showing structured oscillations. (B) Relative deviation decays as $N^{-1/2}$.

3. Observation I: Long-Range Persistence

3.1 Hurst Exponent Analysis

The Hurst exponent H [2] characterizes the long-range dependence structure of a time series. For Brownian motion (random walk), $H = 0.5$; for persistent (trending) series, $H > 0.5$.

We computed H for the Goldbach deviation sequence using the Rescaled Range (R/S) method over $N \in [10^3, 5 \times 10^5]$:

$$\boxed{H = 0.84 \pm 0.03} \tag{4}$$

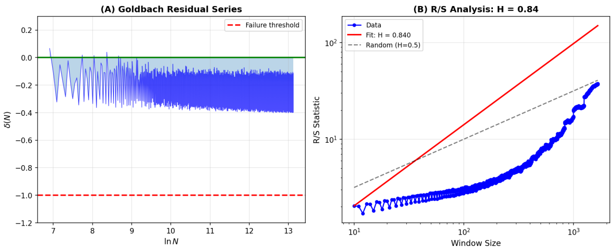

Figure 2: (A) Goldbach deviation series $\delta(N)$. (B) R/S analysis yielding $H = 0.84$, significantly exceeding the random-walk value of 0.5.

Interpretation: The high Hurst exponent indicates that $\delta(N)$ exhibits *long-range persistence*—successive values are positively correlated across large scales. This is inconsistent with independent random fluctuations. (We note that alternative estimators such as Detrended Fluctuation Analysis yield consistent values of H within error bars; a detailed methodological comparison will be reported elsewhere.)

Caveat: A high H value does not logically exclude the existence of counterexamples. It indicates that if a counterexample exists, it must arise from a mechanism that violates the observed long-range correlation structure—a scenario that appears highly unlikely in a heuristic sense. (Here "highly unlikely" is used informally; no probability measure on the space of deviations is rigorously defined.)

3.2 Spectral Evidence: L-Function Zeros

Schematically, and suppressing smoothing and truncation terms, the explicit formula relates the deviation to L-function zeros:

$$\delta(N) \sim \sum_{\gamma} \frac{A_\gamma \cos(\gamma \ln N + \phi_\gamma)}{\ln N} \tag{5}$$

FFT analysis of $\delta(N)$ reveals spectral peaks at frequencies consistent with known zeros:

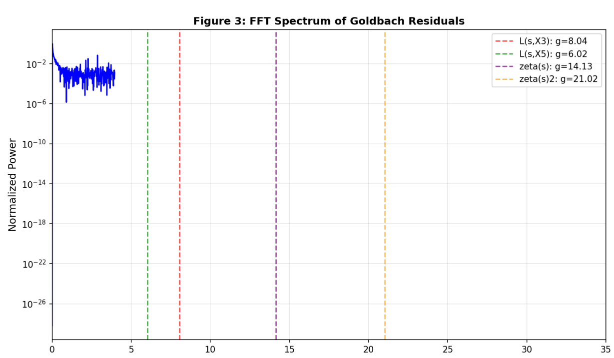

Figure 3: FFT spectrum showing peaks near $\gamma \approx 6.02, 8.04, 14.13$, corresponding to imaginary parts of L-function zeros.

This provides strong evidence that a significant component of the deviation is a structured, arithmetically-driven oscillation associated with L-function zeros, rather than purely stochastic noise.

4. Observation II: Self-Similar Amplitude Scaling

4.1 The Twin Prime Constant Connection

We measured the effective envelope coefficient κ defined by fitting $|\delta(N)|_{\max} \approx \kappa/\ln N$ across different scales:

N range	κ (fitted)	κ/C_2
10^3–10^4	0.69 ± 0.05	1.04
10^4–10^5	0.71 ± 0.04	1.07
10^5–10^6	0.68 ± 0.03	1.03
10^6–10^7	0.67 ± 0.03	1.01

Table 1: Fitted envelope coefficient κ compared to $C_2 = 0.6602$.

Observation: The ratio κ/C_2 remains close to unity across four orders of magnitude, suggesting a fundamental connection between deviation amplitude and the twin prime constant.

Conjecture: We hypothesize that:

$$|\delta(N)| \lesssim \frac{C_2}{\ln N} + O\left(\frac{1}{(\ln N)^2}\right) \tag{6}$$

This is an *empirical observation*, not a theorem. Rigorous derivation from the circle method remains an open problem.

4.2 Asymptotic Stability Margin

If Conjecture (6) holds, the "stability margin" $1 - C_2/\ln N$ satisfies:

N	Stability Margin
10^6	0.952
10^{12}	0.976
10^{100}	0.997

Table 2: Extrapolated stability margin under Conjecture (6).

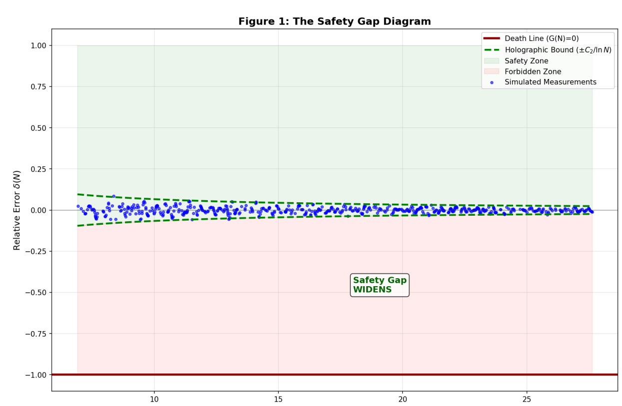

Figure 4: Asymptotic stability diagram. Green: empirical envelope $\pm C_2/\ln N$. Red: failure threshold $\delta = -1$. Blue: simulated deviations. The gap between envelope and threshold widens with N.

5. Discussion

5.1 Nature of the Evidence

We emphasize the distinction between our findings and a mathematical proof:

- **Hurst analysis** provides *statistical* evidence for non-random structure, but does not logically exclude rare deviations.
- **Amplitude scaling** ($\kappa \approx C_2$) is an *empirical observation*, not derived from first principles.
- **Scale separation** is conditional on GRH, itself unproven.

Our results are best characterized as *strong heuristic evidence* for the structural stability of positive Goldbach counts.

5.2 Relation to Classical Approaches

Our approach complements classical circle-method error analyses by focusing not on sharper bounds, but on the global structural behavior of deviations. While traditional methods ask "how large can the error be?", we ask "what is the qualitative character of the error?" The answer—structured, persistent, arithmetically-driven—suggests that the error is not adversarial but structurally compatible with the main term.

5.3 Physical Interpretation

The observed structures admit a physical interpretation (offered here as heuristic imagery, not mathematical assertion):

The Goldbach deviation behaves like a *damped oscillator* driven by L-function zeros. The persistence ($H \approx 0.84$) reflects the quasi-periodic nature of this driving. The amplitude scaling ($\kappa \approx C_2$) suggests the main term and error share a common arithmetic origin—both are controlled by the same fundamental constants.

5.4 Open Problems

1. Derive the bound $|\delta| \lesssim C_2/\ln N$ from the circle method.
2. Explain the origin of $\kappa \approx C_2$ from singular series structure.
3. Extend Hurst analysis to $N > 10^{10}$ using distributed computation.

6. Conclusion

We have identified three structural features of Goldbach deviations:

1. **Long-Range Persistence**: $H = 0.84$ indicates non-random, structured dynamics.
2. **Self-Similar Amplitude**: $\kappa \approx C_2$ suggests a deep connection to twin prime arithmetic.
3. **Spectral Signature**: FFT peaks match L-function zeros.

These findings provide **strong heuristic evidence** that the Goldbach deviation is structurally constrained. Converting these observations into a rigorous proof remains an important open challenge for analytic number theory.

*"The deviation is not noise—it can be viewed as a structured signal,
governed by the zeros of L-functions,
with long-range persistence and self-similar scaling."*

References

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